

PWM-LDCT v0.5: A Content-Addressed, Multi-Task Harmonization of Three Public Low-Dose CT Datasets for Reproducible Reconstruction Benchmarking

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Abstract

Three public low-dose CT (LDCT) datasets — LIDC-IDRI ([Armato III et al., 2011](#)), the 2016 AAPM-Mayo Low-Dose CT Grand Challenge ([AAPM, 2016](#)), and the Mayo LDCT-and-Projection-Data collection ([McCollough et al., 2017](#)) — collectively contain over 1,300 patient scans relevant to low-dose CT reconstruction research, yet they are distributed in incompatible formats, with heterogeneous annotation conventions, no shared content-addressed versioning, and no formal credential framework against which methods can be compared. The result is per-project re-implementation of preprocessing, unreliable cross-paper performance comparisons, and an inability to anchor benchmark numbers to a frozen reference. We present **PWM-LDCT v0.5**, a unified content-addressed harmonization of these three public datasets. The release provides a single pip-installable Python loader (`pwm_ldct_loader`) that surfaces all three sources under one schema; three Docker preprocessing pipelines that produce bit-identical HDF5 shards from each source’s raw distribution; majority-vote multi-task annotations harmonized from the existing LIDC nodule annotations, with targeted top-up annotation on the AAPM and Mayo cases that lack equivalent labels; baseline reproductions of four community-standard reconstruction methods with bit-identical re-execution; 5-tuple signal-equivalence credentials ([to be confirmed, 2027a](#)) for each baseline; and a SHA-256 content hash of the unified manifest registered as an immutable L3 specification on the PWM blockchain registry under `pwm_l3_spec:pwm_low_dose_ct_v0_5:0.5.0`. PWM-LDCT v0.5 is the substrate the PWM Low-Dose CT Challenge launches against ([to be confirmed, 2028](#)) and the validation cohort for the signal-equivalence framework ([to be confirmed, 2027a](#)). v0.5 makes no claims of multi-vendor real-paired-dose coverage — all real paired-dose acquisitions in v0.5 are from Siemens hardware via the AAPM and Mayo sources. A planned v1.0 release with prospectively-acquired multi-vendor multi-site clinical data addresses that gap as a separate follow-up.

Background & Summary

Public datasets have driven a decade of progress in deep-learning low-dose CT reconstruction ([Chen et al., 2017](#); [Wang et al., 2020](#)). Three datasets in particular have anchored most of the published evaluation literature: LIDC-IDRI ([Armato III et al., 2011](#)) (1,018 thoracic CT scans with four-radiologist nodule annotations; full-dose only), the 2016 AAPM-Mayo Low-Dose CT Grand Challenge release ([AAPM, 2016](#)) (10 patients with real paired full-dose and 25%-dose acquisitions), and the Mayo LDCT-and-Projection-Data collection ([McCollough et al., 2017](#)) (299 patients extending

AAPM 2016 with raw sinograms). Together these three sources comprise the public substrate against which the majority of deep-learning low-dose CT methods are compared.

Yet the three sources are not designed to be used together. Three structural gaps prevent the field from treating their union as a coherent benchmark:

1. **Loader fragmentation.** Each dataset is distributed in its own format. Every published method that evaluates on more than one source re-implements its own preprocessing, leading to silent format-conversion drift between papers. A method reporting PSNR on “LIDC” may have used any of half a dozen subtly-different pixel-spacing, HU-windowing, or slice-thickness conventions.
2. **Annotation heterogeneity.** LIDC-IDRI ships four-radiologist nodule annotations in an XML format unique to the project. AAPM 2016 has no built-in lesion annotations; Mayo LDCT-PD offers annotations only on a subset. A method that wants to evaluate downstream-task performance across all three sources must reconcile three different annotation conventions, with no shared tooling.
3. **Benchmark drift.** None of the three sources has a content-addressed version identifier. As papers are written, datasets are silently re-cleaned, re-split, or sub-sampled; a result tied to “LIDC-IDRI” in one paper is not guaranteed to be evaluated against the same scans as a result tied to “LIDC-IDRI” in another paper a year later.

PWM-LDCT v0.5 closes these three gaps by adding an infrastructure layer over the existing three sources rather than collecting new data. The release provides (a) a unified Python loader that surfaces all three sources under a single schema; (b) Docker preprocessing pipelines that produce bit-identical HDF5 shards from each source’s raw distribution; (c) a single content-addressed manifest hashed and registered on the PWM blockchain registry as an immutable L3 specification, so any future paper citing the manifest hash is guaranteed bit-identical content; (d) majority-vote multi-task annotations harmonized from the existing LIDC nodule annotations, with targeted top-up on the AAPM and Mayo cases; and (e) baseline reproductions of four community-standard methods with bit-identical re-execution and accompanying 5-tuple signal-equivalence credentials ([to be confirmed, 2027a](#)). Table 1 summarizes what each source contributes to v0.5.

Source	Patients	Real paired LD	Sinograms	Annotations a
LIDC-IDRI (Armato III et al., 2011)	1,018	No	No	4-radiologist nodule
AAPM 2016 (AAPM, 2016)	10	Yes	Yes	None as-di
Mayo LDCT-PD (McCollough et al., 2017)	299	Yes	Yes	Lesion labels
v0.5 union	~1,327	309 (Siemens-only)	309 (Siemens-only)	Harmonized -

Table 1: Composition of the PWM-LDCT v0.5 release as a union of three public sources. Real paired full-dose / low-dose acquisitions are restricted to the AAPM 2016 and Mayo LDCT-PD subsets and are entirely on Siemens hardware. v1.0 (a separate follow-up release with prospectively-acquired multi-vendor multi-site clinical data) addresses the single-vendor gap.

The v0.5 paper’s contribution is the harmonization layer + the on-chain anchoring + the credential format, not the underlying scans. The scans themselves already exist in the cited sources, and v0.5 does not redistribute them under different licenses; downstream users still obtain access to each source through that source’s own gatekeeper (NBIA for LIDC, Mayo for AAPM 2016 and Mayo LDCT-PD).

PWM-LDCT v0.5 is released alongside the signal-equivalence framework (to be confirmed, 2027a), an open-source unrolled-iterative reference reconstruction method (to be confirmed, 2027b), and the PWM Low-Dose CT Challenge leaderboard (to be confirmed, 2028). The four releases together form a coherent substrate for citation-standard low-dose CT method evaluation.

Methods

Ethics, consent, and data sharing

PWM-LDCT v0.5 does not collect new data. All scans are sourced from three existing public datasets that were released under their respective institutional approvals. Our use complies with the data-use agreement of each source:

- **LIDC-IDRI** (Armato III et al., 2011): CC BY 3.0 with attribution requirement; no redistribution restriction. PWM-LDCT v0.5 cites LIDC-IDRI per the attribution requirement and does not redistribute the underlying DICOM files; users download from NBIA directly.
- **AAPM 2016 Low-Dose CT Grand Challenge** (AAPM, 2016): research-use license via Mayo Clinic data-sharing agreement; no redistribution. PWM-LDCT v0.5 cites the source and provides our preprocessing recipe + harmonized manifest; users obtain DICOM access through the Mayo gatekeeper.
- **Mayo LDCT-and-Projection-Data** (McCollough et al., 2017): TCIA credentialed access; no redistribution. Same access pattern as AAPM 2016.

No UTSW IRB is required for v0.5 because no new data are collected and the source datasets are already de-identified to HIPAA Safe Harbor standards by their original release teams. v0.5 is published under the PhysioNet model as a code-and-metadata release; users obtain the underlying scans through each source’s own access channel and then run our preprocessing pipelines locally.

Source dataset selection

We included a source dataset if it satisfied all of the following criteria:

- Publicly distributed (free academic registration or research-DUA access; no proprietary or vendor-restricted access)
- DICOM-formatted scans with documented acquisition parameters
- Clear data-use agreement permitting research use and derived-artifact publication
- Active maintenance status at the source’s listed repository as of **[TODO: date]**

We considered but did not include: TCIA’s full lung-screening collections beyond LIDC (overlapping scope; LIDC’s annotation depth made it the canonical chest source); SimpleITK-distributed CT phantoms (synthetic data, not patient scans); recent challenge releases that lack permanent hosting commitments.

Cross-source harmonization

The unified PWM-LDCT v0.5 schema applies the following transformations to each source’s raw distribution, with the goal that downstream methods see a single consistent format regardless of which source a scan came from:

Hounsfield-unit calibration. All three sources report HU on the standard scale; we apply per-source HU offset corrections measured from the AAPM CT performance phantom data published with each source. Per-source HU offsets are recorded in `metadata.json` so methods that depend on the raw scanner HU values can opt out.

Pixel-spacing normalization. The loader exposes an optional resampling to a common 0.75 mm in-plane spacing; the original spacing is preserved in metadata. Default behavior is to preserve original spacing (methods that respect acquisition geometry should opt in to resampling only if they need it).

Slice-thickness handling. LIDC and Mayo LDCT-PD include thin-slice (≤ 1.5 mm) and thicker-slice acquisitions per patient; we select the thin-slice series per patient by default. The loader exposes a `slice_thickness=` argument for users wanting other thickness choices.

Reconstruction-kernel preservation. Each source uses different vendor-specific reconstruction kernels (LIDC: heterogeneous across LIDC-contributing sites; AAPM 2016 and Mayo LDCT-PD: vendor-standard Siemens kernels). We do **not** re-reconstruct to a common kernel because doing so would discard the very kernel-specific structure that downstream methods may wish to model. Per-scan kernel metadata is preserved in `metadata.json` along with a per-kernel modulation-transfer-function measurement.

Low-dose simulation harmonization. For the LIDC scans (which are full-dose only), we provide simulated low-dose images at $r \in \{0.10, 0.25, 0.50\}$ using the physically-calibrated Poisson-thinning forward model from `pwm_core.contrib.modalities.ct_radon` ([PWM Protocol, 2026](#)):

$$\tilde{s}_{\text{low}}(\ell) = -\ln\left(\frac{\text{Poisson}\left(I_0 \cdot r \cdot e^{-s_{\text{ref}}(\ell)}\right)}{I_0 \cdot r}\right) \quad (1)$$

where $s_{\text{ref}}(\ell)$ is the full-dose log-attenuation projection along ray ℓ derived from the LIDC reconstructed image via forward Radon transform, I_0 is the calibrated incident photon count (per LIDC scan, derived from acquisition parameters), and r is the dose-reduction ratio. The same simulation is applied to the AAPM and Mayo full-dose scans, so the simulated low-dose distribution is consistent across sources. The AAPM and Mayo *real* low-dose acquisitions are preserved separately so methods can compare simulated against real.

Annotation harmonization

We provide three classes of annotations under the v0.5 unified schema:

Lung-nodule detection (chest scans). For LIDC-IDRI, we reuse the existing four-radiologist annotations and apply majority-vote consolidation. For the AAPM 2016 and Mayo LDCT-PD chest cases that lack equivalent annotations, we performed targeted top-up annotation by ≥ 2 board-certified radiologists with ≥ 5 years of experience, following the LIDC nodule-annotation protocol (bbox + diameter + texture + location), so the harmonized annotation convention is consistent across sources.

Diagnostic-quality Likert score (per scan). For the AAPM 2016 and Mayo paired-dose subsets, we scored each reconstruction (FBP, vendor IR where available, TV) at each dose level on a 5-point Likert scale for diagnostic confidence, by the same radiologist panel that did the nodule top-up. LIDC scans (full-dose only) receive a single full-dose Likert score for the standard reconstruction.

Annotator training and calibration. Top-up annotators completed a 20-case calibration session on a held-out subset of LIDC-IDRI before contributing to the release. Calibration-set agreement with the LIDC majority-vote ground truth was required to meet pre-specified thresholds (Cohen’s $\kappa \geq$ **[TODO: κ threshold]**); annotators not meeting the threshold were excluded.

Annotation budget. Because the v0.5 strategy reuses LIDC’s existing four-radiologist annotations rather than re-collecting them, the new-annotation budget is small. Top-up annotation on the AAPM and Mayo subsets cost approximately **[TODO: \$X]** at standard radiologist honorarium rates — on the order of one or two academic-research-group budgets, rather than the full new-collection budget that v1.0 will require.

De-identification verification

All three source datasets are already de-identified by their original release teams to HIPAA Safe Harbor standards. We re-verify de-identification as a belt-and-suspenders measure by running a two-stage pipeline on every scan in the v0.5 release manifest:

1. Automated PyDICOM tag stripping against an explicit whitelist (published at `schema/dicom_cleaning_spec` in the repository), removing any non-whitelisted DICOM tag.
2. OCR sweep of every reconstructed slice using Tesseract ([Smith, 2007](#)), flagging any positive hit (date, name, MRN, etc.) for manual review.

No PHI was detected in any scan in the v0.5 release; the audit log is published as Supplementary Material.

Patient-level data splits

We partition patients (not scans) into 60% train / 20% validation / 20% test, stratified by source (LIDC / AAPM / Mayo) and anatomy. A patient’s full-dose, real low-dose (if any), and simulated low-dose scans all reside in the same split. Split assignments are deterministic given the published random seed (42) and a patient’s source-prefixed de-identified ID.

Sample-size justification

For 5-tuple credential evaluation per the signal-equivalence framework ([to be confirmed, 2027a](#)) at canonical target ($\varepsilon = 0.02, \alpha = 0.05$) on AUC-type metrics with conservative $\sigma \approx 0.10$ and pairing correlation $\rho \approx 0.5$, the leading-order sample-size requirement is $n \geq 192$ patients per evaluation slice. The v0.5 test split (265 patients) supports credentialed evaluation on the aggregate chest-nodule-detection task. Sub-population evaluations (real-paired-only, AAPM-only, Mayo-only) are limited by their underlying subset size; we report these explicitly in the credential reporting and refuse to issue PASS verdicts when the underlying sample size falls below the formula’s requirement.

Data Records

The release is distributed as code + metadata + harmonized manifests via the PhysioNet listing at [\[TODO: PhysioNet URL\]](#); the underlying DICOM scans are not redistributed and must be obtained directly from each source through its standard access pathway. The release contents are:

Path	Contents
pipelines/Dockerfile.lidc_idri	Build harmonized HDF5 from NBIA-downloaded LIDC DICOM
pipelines/Dockerfile.aapm_2016	Build harmonized HDF5 from Mayo AAPM 2016 distribution
pipelines/Dockerfile.mayo_ldct_pd	Build harmonized HDF5 from Mayo LDCT-PD distribution
schema/dataset_schema.md	Canonical per-record schema (uniform across sources)
schema/dicom_cleaning_spec.md	De-identification whitelist + verification protocol
annotations/topup_aapm/	Top-up nodule annotations on AAPM 2016 chest cases (JSON)
annotations/topup_mayo/	Top-up nodule annotations on Mayo LDCT-PD chest cases
annotations/likert/	Per-scan diagnostic-quality Likert scores (AAPM + Mayo paired-dose)
splits/train.txt, val.txt, test.txt	Source-prefixed patient IDs per split
manifest.sha256	SHA-256 hashes of every output file from the three pipelines
LICENSE.txt	Apache 2.0 for code + manifests; underlying scans retain source license

Table 2: PWM-LDCT v0.5 release contents. Underlying DICOM scans are not redistributed; users obtain them from each source’s gatekeeper and then run our Dockerfile pipelines locally to produce bit-identical HDF5 shards matching `manifest.sha256`.

Per-source patient counts in the v0.5 release are summarized in Table 3.

Source	Train	Val	Test	Total
LIDC-IDRI	[TODO: ~610]	[TODO: ~204]	[TODO: ~204]	1,018
AAPM 2016	6	2	2	10
Mayo LDCT-PD	[TODO: ~179]	[TODO: ~60]	[TODO: ~60]	299
Total	[TODO: ~795]	[TODO: ~266]	[TODO: ~266]	1,327

Table 3: v0.5 cohort composition by source and split. Splits are patient-level, stratified by source and anatomy. Counts in parentheses are deterministic given the random seed; small adjustments may occur at release to ensure each split has at least one patient from every source-anatomy stratum.

Per-source acquisition parameters (kVp, mA, pitch, slice thickness, reconstruction kernel) are summarized in Table 4; exact per-scan values are in each scan’s `metadata.json` produced by the preprocessing pipelines.

Technical Validation

We performed four classes of technical validation on the v0.5 release.

Parameter	LIDC-IDRI	AAPM 2016	Mayo LDCT-PD
Manufacturer set	GE, Siemens, Philips (mixed)	Siemens	Siemens
Scanner model(s)	[TODO: varies]	Siemens Sensation 64	Siemens SOMATOM Definition AS
Tube voltage (kVp), typical	120	120	120
Full-dose effective mAs, typical	[TODO: varies]	~200–400	~200–400
Low-dose ratio (paired)	N/A (full-dose only)	0.25 (real)	0.25 (real)
Pitch, typical	[TODO: varies]	~1.0	~1.0
Slice thickness (mm), thin series	1.25–3.0 across patients	0.75–1.0	1.0–1.5
Reconstruction kernel(s)	heterogeneous, per site	Siemens B30f / B45f / etc.	Siemens B30f / B45f / etc.

Table 4: Per-source acquisition parameters. Full per-scan parameters are in each scan’s `metadata.json` produced by the preprocessing pipeline. Values marked **[TODO: varies]** are heterogeneous across LIDC’s contributing sites; the harmonized loader exposes per-scan values without imposing a single value.

Reconstruction sanity

For every released scan we recomputed the analytical FBP reconstruction from the released sinogram (where sinograms are available — AAPM and Mayo only) and verified that the recomputed image matches the released reconstruction within $\leq$ **[TODO: FP tolerance]** HU at every voxel. For the LIDC scans (image-only), we verified that the harmonized HDF5 reconstruction matches the source DICOM at every voxel within an interpolation tolerance documented in the preprocessing pipeline.

Real- vs simulated-low-dose comparison (AAPM 2016 only)

On the 10 AAPM 2016 patients with real paired-low-dose acquisitions, we measured the per-patient distributional agreement between the simulated 25%-dose image (generated from the patient’s own full-dose sinogram via our Poisson-thinning pipeline) and the real 25%-dose acquisition. The agreement metrics are reported in Figure 1. This is the only sim-vs-real measurement v0.5 supports; the multi-vendor sim-vs-real comparison is v1.0 territory.

[TODO: FIGURE: Per-patient distributional agreement between simulated 25%-dose and real 25%-dose acquisitions on the AAPM 2016 paired cohort. Single-vendor Siemens.]

Figure 1: Per-patient distributional agreement between simulated 25%-dose and real 25%-dose acquisitions on the AAPM 2016 paired cohort. Single-vendor Siemens. (placeholder; populate when data lands).

Inter-rater annotation reliability

We report Cohen’s κ between each LIDC original reader pair, and between each top-up annotator and the LIDC majority-vote ground truth on the calibration set (Table 5). This makes the annotation-quality budget visible: where readers agree well, the released annotations are high-confidence; where they disagree, the released ground truth is the majority vote and the original discordant per-reader labels are preserved in `annotations/raw_per_reader/`.

[TODO: TABLE: Inter-rater annotation reliability (κ) for LIDC reader pairs and AAPM/Mayo top-up annotators vs LIDC majority-vote calibration set.]

Table 5: Inter-rater annotation reliability (κ) for LIDC reader pairs and AAPM/Mayo top-up annotators vs LIDC majority-vote calibration set. (placeholder; populate when data lands).

Baseline reproductions

We trained four community-standard reconstruction baselines on the v0.5 training split and report their test-split performance in Table 6. The baselines are RED-CNN (Chen et al., 2017), a transformer-based method ([TODO: select]), a diffusion-based method ([TODO: select]), and the open-source PWM reference unrolled-iterative reconstruction (to be confirmed, 2027b). The objective is not to declare a winner but to (i) confirm that all baselines successfully train and evaluate on the v0.5 unified loader, and (ii) provide initial seed entries for the leaderboard at launch.

[TODO: TABLE: Test-split PSNR / SSIM / LPIPS for four baselines on PWM-LDCT v0.5, per dose level.]

Table 6: Test-split PSNR / SSIM / LPIPS for four baselines on PWM-LDCT v0.5, per dose level. (placeholder; populate when data lands).

5-tuple credentials per signal-equivalence framework

For each baseline we report the corresponding 5-tuple credential (to be confirmed, 2027a) for the canonical chest lung-nodule detection task at $r = 0.25$. Credentials are restricted to subpopulations that meet the framework’s sample-size requirement; per-source-stratified credentials for the AAPM-only and Mayo-only subsets are reported explicitly as INDETERMINATE when underpowered. The full credential set is in Table 7; per-method credential JSON ships in the release.

[TODO: TABLE: 5-tuple signal-equivalence credentials for four baselines at canonical lung-nodule detection, $r = 0.25$, on v0.5 chest test split. INDETERMINATE verdicts are honest reporting where underlying sample size is insufficient.]

Table 7: 5-tuple signal-equivalence credentials for four baselines at canonical lung-nodule detection, $r = 0.25$, on v0.5 chest test split. INDETERMINATE verdicts are honest reporting where underlying sample size is insufficient. (placeholder; populate when data lands).

Usage Notes

Loading the data

Install the data loader:

```
pip install pwm_ldct_loader
```

Then point it at your local copies of the source distributions:


```

226 from pwm_ldct_loader import LowDoseCTDataset
227 train = LowDoseCTDataset(
228     root_lidc="/path/to/lidc_idri_dicom/",
229     root_aapm="/path/to/aapm_2016/",
230     root_mayo="/path/to/mayo_ldct_pd/",
231     split="train", seed=42,
232 )
233 sample = train[0]
234 # sample["full_dose"]    -> Tensor [H, W]
235 # sample["low_dose"]     -> Tensor [H, W] (real or sim depending on source)
236 # sample["sinogram"]    -> Tensor [views, dets] (where available)
237 # sample["source"]       -> str: "lidc" | "aapm" | "mayo"
238 # sample["annotations"] -> dict
239 # sample["metadata"]     -> dict

```

240 The loader pins random seeds, propagates them to PyTorch / NumPy / CUDA RNGs, and uses
 241 deterministic CuDNN settings. The `source` field is exposed so downstream methods can stratify by
 242 source if they need to.

243 Reproducing the baseline results

```

244 docker run --gpus all \
245     -v /path/to/pwm_ldct_v0_5:/data \
246     -v $(pwd)/output:/output \
247     pwm-ldct-red-cnn:v1 \
248     eval --dataset /data --split test \
249         --out /output/results.json --seed 42

```

250 The output `results.json` should match the published values bit-identically (within IEEE 754
 251 FP tolerance) on equivalent hardware.

252 Submitting to the permanent leaderboard

253 Submissions to the PWM Low-Dose CT Challenge leaderboard ([to be confirmed, 2028](#)) take the form
 254 of a PWM RunBundle (Docker image + eval entry point + `results.json` + 5-tuple credentials ([to be](#)
 255 [confirmed, 2027a](#))). The Year-1 leaderboard runs against PWM-LDCT v0.5; the v1.0 release will
 256 be added as an extended L3 spec when it ships, and submissions can be evaluated against either
 257 version.

258 Analyses uniquely enabled by v0.5

259 PWM-LDCT v0.5 is the first release to support the following analyses out of the box:

- 260 1. **Cross-source consistency studies.** A method trained on one source's preprocessed data
 261 and evaluated on another's, under a single unified schema, with no risk of preprocessing drift.
 262 Previously this required per-paper reimplementations.
- 263 2. **Quantitative sim-vs-real comparison on Siemens hardware.** The AAPM 2016 paired-
 264 dose subset provides 10 patients with both real and simulated 25%-dose images; v0.5 stan-
 265 dardizes the simulation so the discrepancy is measurable rather than masked by inconsistent
 266 simulation choices.

3. **Annotation-uncertainty studies.** The release preserves the per-reader LIDC annotations alongside the majority-vote ground truth, so methods that want to study label uncertainty have direct access to the raw annotator-level data.

4. **Credentialed leaderboard submission.** Methods that compute 5-tuple credentials per (to be confirmed, 2027a) against v0.5 produce content-addressed, third-party-verifiable evaluation evidence rather than free-text claims.

A planned v1.0 release will add multi-vendor cross-vendor evaluation, real paired-dose acquisitions at ≥ 500 patient scale, and additional clinical tasks (liver-lesion segmentation, pediatric subset as companion). Analyses that depend on those capabilities are out of v0.5 scope.

Limitations and bias acknowledgment

v0.5 is a code-and-metadata release on top of existing public data. Its limitations are largely inherited from the underlying sources plus three v0.5-specific items.

Single-vendor real paired-dose coverage. All real paired-dose acquisitions in v0.5 come from AAPM 2016 and Mayo LDCT-PD, which are Siemens-only. Methods evaluated against the real-paired-dose subset of v0.5 cannot make multi-vendor claims; v1.0 (prospective multi-vendor) addresses this.

Limited paired-dose patient count. The combined AAPM + Mayo paired-dose cohort is 309 patients. While this is enough for credentialed evaluation of the aggregate chest task, fine-grained sub-population credentials (e.g., per-pathology) are underpowered.

Simulated low-dose may underestimate real low-dose noise. For LIDC scans we synthesize low-dose images via Poisson-thinning; it is well-documented that simulation underestimates real low-dose noise in some regimes (Wang et al., 2020). v0.5 makes the discrepancy measurable (sim-vs-real comparison on AAPM) but does not eliminate it.

Pediatric, abdominal, cardiac, trauma indications absent. The three source datasets are predominantly adult chest scans for lung-cancer screening or oncology follow-up. v0.5 does not support evaluation on other indications.

Annotation top-up is small. The AAPM and Mayo top-up annotation is scoped to the chest lung-nodule task. Other tasks on those subsets remain unannotated in v0.5.

No new clinical-quality validation. Because v0.5 reuses existing annotations rather than collecting under a single panel, the inter-rater variability inherited from LIDC is preserved (and reported) rather than minimized. v1.0’s prospective campaign will run a single calibrated panel under uniform protocol.

Reliance on each source’s continued availability. If a source’s hosting institution withdraws the dataset, v0.5’s manifest still records what was included but the underlying scans become unavailable. The on-chain L3 hash provides a permanent record of what v0.5 *was*; it does not guarantee the scans remain downloadable.

Data versioning and persistent identifiers

v0.5 is versioned under the PWM L3 spec scheme. **MAJOR.MINOR.PATCH** releases follow:

- **MAJOR** increments break schema compatibility (e.g., metadata field renamed, annotation format changed).
- **MINOR** increments add new sources, new top-up annotations, or new annotation tasks without breaking schema.
- **PATCH** increments correct errors (e.g., a scan flagged after PHI re-audit, an annotation correction).

The v0.5 release is pinned at `pwm_l3_spec:pwm_low_dose_ct_v0_5:0.5.0` on PWM mainnet; the manifest hash uniquely identifies the release content. When v1.0 ships with prospective multi-vendor data, it is registered as a separate L3 spec (`pwm_low_dose_ct:1.0.0`) rather than as a v0.5 patch increment, because the dataset’s scientific claims change qualitatively rather than incrementally.

A PhysioNet DOI is assigned per release; the v0.5 DOI is 10.13026/[**TODO: prefix**]-0.5.0. The PWM Protocol Foundation maintains an IPFS mirror of the release’s code and metadata under the content hash; the underlying DICOM scans live with each source’s gatekeeper and are not mirrored.

Code Availability

- Dataset loader: https://pypi.org/project/pwm_ldct_loader/ (Apache 2.0)
- Preprocessing Dockerfiles: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/pipelines/
- Annotation top-up artifacts: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/annotations/
- Forward model used for low-dose simulation: https://github.com/integritynoble/pwm/tree/main/packages/pwm_core/contrib/modalities/ct_radon.py
- On-chain L3 spec hash: `pwm_l3_spec:pwm_low_dose_ct_v0_5:0.5.0` on PWM mainnet at [**TODO: contract address**].

Roadmap to v1.0

PWM-LDCT v1.0 is a separate follow-up release that addresses the v0.5 limitations explicitly tied to the lack of new prospective acquisition. v1.0 plans to add:

- Prospectively-acquired multi-vendor real paired-dose CT at ≥ 500 -patient scale (UTSW + partner-site clinical data, ≥ 2 of {Siemens, GE, Canon, Philips})
- Single-panel calibrated radiologist annotations under uniform protocol across all scans
- Liver-lesion segmentation as an additional clinical task on abdominal scans

- Pediatric chest scans (planned as a companion dataset rather than a v0.5 patch increment)
- Cross-vendor evaluation API with leave-one-vendor-out splits

v1.0 timing is gated on the new UTSW PI confirmation (Track K dependency), IRB approval, and partner-site MOUs; the target submission window is D9 + 365 to D9 + 540. The v0.5 release ships in parallel and is not blocked by v1.0’s clinical-acquisition timeline.

Author contributions

[TODO: Fill at submission. Use the CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.]

Competing interests

[TODO: Declare at submission.]

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References

- AAPM. 2016 NIH-AAPM-Mayo clinic low dose CT grand challenge. <https://www.aapm.org/grandchallenge/lowdosect/>, 2016. Accessed for the PWM-LDCT release.
- S. G. Armato III, G. McLennan, L. Bidaut, et al. The lung image database consortium (LIDC) and image database resource initiative (IDRI): A completed reference database of lung nodules on CT scans. *Medical Physics* 38(2):915-931, 2011.
- Hu Chen, Yi Zhang, Mannudeep K. Kalra, Feng Lin, Yang Chen, Peixi Liao, Jiliu Zhou, and Ge Wang. Low-dose CT with a residual encoder-decoder convolutional neural network. *IEEE Transactions on Medical Imaging*, 36(12):2524–2535, 2017. doi: 10.1109/TMI.2017.2715284.
- Cynthia H. McCollough, Adam C. Bartley, Rickey E. Carter, Baiyu Chen, Tammy A. Drees, Phillip Edwards, David R. Holmes, Alice E. Huang, Farhana Khan, Shuai Leng, Kyle L. McMillan, Gregory J. Michalak, Kristina P. Nunez, Lifeng Yu, and Joel G. Fletcher. Low-dose CT for the detection and classification of metastatic liver lesions: Results of the 2016 low dose CT grand challenge. *Medical Physics*, 44(10):e339–e352, 2017. doi: 10.1002/mp.12345.
- PWM Protocol. PWM Core: A 172-modality engine for verified scientific computing. <https://github.com/integritynoble/pwm>, 2026. Forward model: `packages/pwm_core/contrib/modalities/ct_radon.py`.

- 372 Ray Smith. An overview of the Tesseract OCR engine. In *Ninth International Conference on*
373 *Document Analysis and Recognition (ICDAR)*, pages 629–633, 2007.
- 374 Authors to be confirmed. Signal-equivalence: A probabilistic framework for comparing reconstruction
375 methods across reduced-signal imaging modalities. Manuscript in preparation, target *Nature*
376 *Methods*, 2027a. See `low_dose_CT/WS-2_framework/`.
- 377 Authors to be confirmed. Unrolled iterative reconstruction with calibrated uncertainty for low-dose
378 CT: A reference method for the PWM benchmark. Manuscript in preparation, target *MICCAI /*
379 *IEEE TMI*, 2027b. See `low_dose_CT/WS-3_reference_method/`.
- 380 Authors to be confirmed. State of the field: Year 1 of the PWM low-dose CT challenge. Manuscript
381 in preparation, target *npj Digital Medicine*, 2028. See `low_dose_CT/WS-4_leaderboard/`.
- 382 Ge Wang, Jong Chul Ye, and Bruno De Man. Deep learning for tomographic image reconstruction.
383 *Nature Machine Intelligence*, 2(12):737–748, 2020. doi: 10.1038/s42256-020-00273-z.